

Q1.

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user. 1. Accept a Linux command as input. 2. Create a child process using **fork()**. 3. Execute the command in the child process using an appropriate **exec()** system call. 4. Allow the parent process to wait for the child using **wait()**. 5. Display the Process ID (PID) of both parent and child processes.

```
venkatesh@LAPTOP-LL9RQ023:~$ cat week1.c

#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <sys/wait.h>

int main() {
    char cmd[10];
    pid_t pid;

    printf("Enter the Command\n");
    scanf("%s", cmd);

    pid = fork();

    if (pid == 0) {
        execlp(cmd, cmd, NULL);
        printf("exec() system call failed\n");
        exit(1);
    } else {
        wait(NULL);
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
    }

    return 0;
}

venkatesh@LAPTOP-LL9RQ023:~$ gcc week1.c
venkatesh@LAPTOP-LL9RQ023:~$ ./a.out
Enter the Command
date
Mon Aug 17 07:02:10 UTC 2026
Parent PID: 711
Child PID: 714
venkatesh@LAPTOP-LL9RQ023:~$ |
```